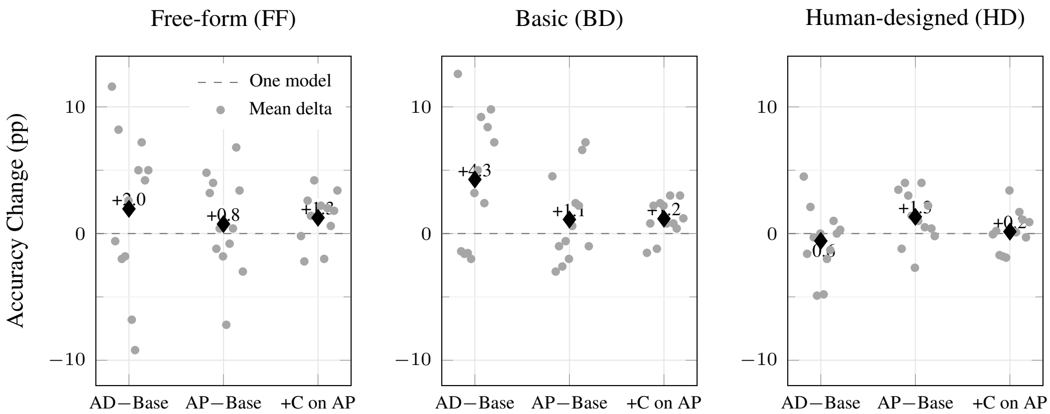

Figure 3: Per-model percentage-point change under symbolic interventions, aggregated from Table A.3. Each gray dot is one model; black diamonds mark mean deltas. 'AD−Base' and 'AP−Base' measure gains over the minimal-context symbolic baseline, while '+C on AP' isolates the incremental change from AP to AP+C. The zero line marks no change. This makes the reviewer-facing pattern explicit: Action Descriptions help most on FF and BD, AP is slightly more stable on HD, and concept conditioning yields modest but heterogeneous model-level effects.

available, a single additional image does not unlock further gains.

5.5 Randomization Controls: Sensitivity to Structure

To test whether models rely on structured rule induction rather than token-level heuristics, we introduce two perturbations in the Action Program regime.

In *Category Permutation*, we randomly reassign support examples between positive and negative sets within each problem, while keeping the query and its gold label fixed. This manipulation preserves token inventories but breaks the original support partition. Performance remains above 50% on both BD and FF, which is expected under this protocol because the unchanged query can still match alternative regularities induced by the shuffled support sets; the relevant diagnostic is the drop relative to the unperturbed AP condition. Under that comparison, FF still drops by about 7 points and BD by about 9 points (Table 1).

In *Sequence Permutation*, we shuffle the order of drawing actions in the query program while leaving the support examples, query image, and gold label intact. BD accuracy drops by about 5 points relative to AP, but FF drops by about 20 points, showing that ordered stroke composition matters most in the Free-form regime.

The two perturbations affect Basic and Free-form tasks in systematically different ways. On Basic (BD) problems, shuffling the assignment of support programs to positive vs. negative sets reduces pooled C–G AP accuracy from 68.8% to 60.1% (–8.7 points), whereas permuting the action sequence in the query program yields a smaller drop to 64.2% (–4.6 points), suggesting that coherent support-set structure is more critical than exact stroke order for these library-based shapes. In contrast, on Free-form (FF) problems, where concepts depend more directly on stroke-level geometry, query sequence permutation is far more damaging: accuracy falls from 78.1% to 58.0% (–20.1 points), compared to 70.6% (–7.5 points) under category permutation (more in Appendix E.6).

6 Discussion

Our results on LOGO reveal a strong dependence of abstract visual reasoning performance on representational format. VLMs operating on raw pixels perform near chance on these Bongard-style tasks, whereas the same underlying reasoning engines, when supplied with structured symbolic input, achieve large gains that, on some categories, approach reported human performance from prior work. This shift, together with the minimal impact of adding a visual anchor and the sharp degradations under structural randomization, highlights a representational bottleneck on Bongard-style tasks: current visual encoders, though successful at object recognition and retrieval, fail to capture the precise geometric and relational structure required for Bongard-style abstraction.

The Componential–Grammatical (C–G)

paradigm isolates this representational gap. Symbolic input acts as a grounding scaffold: it preserves object identity, shape structure, and compositional relations that are typically lost or blurred in visual embeddings. Importantly, these representations are interpretable and amenable to structured manipulation, echoing earlier proposals to treat Bongard problems as inference in a visual language of primitives and relations (Depeweg et al., 2024; Grinberg et al., 2023), and our grounded C–G experiments show that, within the VLM-capable subset, once such a scaffold is in place additional visual evidence offers at best marginal benefits.

We also observe that input format interacts with task complexity. Natural-language descriptions offer gains on simple categorization tasks (e.g., "contains a triangle") but degrade on more abstract tasks that require global structure (e.g., symmetry or convexity). Procedural grammars, by contrast, support more reliable reasoning on complex problems. The perturbation results mirror this pattern: BD accuracy is more sensitive to shuffling which programs belong to the positive vs. negative sets, whereas FF accuracy collapses primarily when we disrupt the ordered composition of strokes in the query program, indicating that different concept classes rely on different facets of symbolic structure. These format-specific effects suggest that there is no universal "best" representation; alignment between representational form and task structure is critical. Model behavior varies as well. Larger models show strong relative gains from symbolic input but are sensitive to verbosity and format. Smaller, instruction-tuned models can sometimes underperform when exposed to over-structured prompts, highlighting brittleness to task framing (more in Appendix E.4). Overall, symbolic inputs help most when they preserve structure without overloading the model with linguistic noise, and the randomization controls indicate that it is precisely this relational and compositional structure—rather than superficial token frequencies—that models rely on, especially for stroke-level Free-form tasks.

These results support a modular view of multimodal AI: instead of relying solely on end-to-end systems, it may be more effective to explicitly separate perception from reasoning, using intermediate symbolic representations as a grounding interface. This design pattern is already common in domains such as formal mathematics and program synthesis, and is increasingly reflected in neurosymbolic visual systems: SBSD models Bongard-LOGO concepts via probabilistic concept models over symbolic shape descriptors (Song and Yuan, 2024), NSA solves ARC tasks by combining neural proposals with search in a domain-specific language (Batorski et al., 2025), agent architectures such as VLAgent plan executable programs from visual input before calling neural modules for execution (Xu et al., 2025), and PRISM explicitly decouples perception and reasoning in VLM pipelines by converting visual information into a textual intermediate representation before downstream inference (Qiao et al., 2024). Our findings extend this pattern to large language models on Bongard-LOGO, suggesting that general-purpose LLMs function effectively as inference engines once grounded in an appropriate visual language. In this sense, C–G should be viewed as a diagnostic upper bound on what current LLMs can do once grounded in an appropriate visual language, rather than as a proposal for solving perception.

Work on compositional visual reasoning in VLMs points to a complementary axis of progress: unrolling reasoning into multi-step procedures. Progressive alignment methods explicitly model multi-granular interactions between language and image regions, and GFlowVLM shows that CoT style multi-step reasoning improves performance on complex visual tasks (Le et al., 2025; Kang et al., 2025). Neurosymbolic agents similarly decompose queries into symbolic subgoals and verify intermediate results (Xu et al., 2025). Our experiments are orthogonal in that, without changing model internals or prompting them for explicit chains of thought, we demonstrate that simply changing the *representation format* — from pixels to LOGO-style programs or descriptions—already unlocks large gains. Together, these lines of work suggest that multi-step reasoning mechanisms and better grounding interfaces are complementary: CoT procedures operate over whatever representations they are given, and our results indicate that providing a suitable visual language is a powerful lever for improving downstream reasoning.

At the same time, our oracle symbolic interface should not be conflated with natural-image reasoning. In natural-image settings, the intermediate representation itself must be inferred from pixels, typically with noise and partial coverage. Recent representation-first pipelines such as Componential Analysis (Vaishnav and Tammet, 2025) and PRISM (Qiao et al., 2024) point toward this direction by first constructing textual or structured world models and then reasoning over them; natural-image Bongard benchmarks such as Bongard-OpenWorld and Bongard-HOI provide testbeds for whether such approximate symbolic interfaces transfer beyond synthetic geometry (Wu et al., 2024; Jiang et al., 2022). Our results therefore best support a conditional claim: if a model is given a faithful symbolic abstraction, its downstream reasoning can be much stronger than its raw-pixel performance suggests.

Finally, recent "Bongard in Wonderland" evaluations show that even frontier foundation models (e.g., GPT-4o) often fail on elementary Bongard concepts such as spirals (Wüst et al., 2024). Alongside our symbolic upper bounds and prior visual-language solvers (Depeweg et al., 2024; Grinberg et al., 2023), this convergence of evidence supports a common conclusion: robust abstract visual reasoning on Bongard-style benchmarks is unlikely to emerge from monolithic end-to-end VLMs alone, and instead appears to require explicit representational structure, whether via hand-engineered operators, probabilistic visual languages, or program-like grounding interfaces for LLMs.

7 Limitations

While the C–G approach reveals useful structure–reasoning interactions, it comes with several limitations (expanded in Appendix F). *First*, Bongard-LOGO is a synthetic benchmark tailored to abstract geometry, which makes it ideal for perturbation studies (grounding, randomization) but leaves open how well our findings about symbolic grounding transfer to natural-image domains such as Bongard-OpenWorld or Bongard-HOI, where the intermediate representation must itself be inferred from cluttered scenes (Wu et al., 2024; Jiang et al., 2022; Vaishnav and Tammet, 2025). *Second*, our setup assumes oracle access to ground-truth action programs, effectively removing perception from the loop; in realistic settings such programs would need to be inferred from pixels via inverse graphics or symbolic detectors, and our grounded C–G variants only partially probe this gap, since they still rely on perfect symbolic inputs and add only a single visual anchor. *Third*, because we bypass perception entirely, C–G is diagnostic rather than prescriptive: it shows what current models can do *given* suitable structure, but not how to learn such representations, pointing to the need for learned front-ends that induce approximate programs or scene graphs from raw data. *Fourth*, our model set, though broad, does not systematically cover architectures with explicit geometric inductive biases, so our observations about capacity and brittleness may not generalize uniformly. Finally, even with perfect symbolic input, models still struggle on Human-designed problems involving global structure and multiple abstraction levels, echoing results from ARC and related benchmarks that reasoning over rich concepts remains challenging even for neurosymbolic systems. Inferring symbolic structure from perception, aligning it with downstream tasks, and scaling robust reasoning over such representations remain important directions for future work.

8 Conclusion

We have used Bongard-LOGO as a controlled testbed to study how representation shapes abstract visual reasoning in large language models. Comparing end-to-end VLMs on images to the same reasoning engines operating on ground-truth symbolic programs suggests that, on this benchmark, a large portion of the performance gap is attributable to perceptual representations rather than to a lack of reasoning capacity. The Componential–Grammatical (C–G) interface, together with grounded and randomization variants indicates that explicit symbolic structure is the main driver of gains in our experiments, while additional visual anchoring and surface-level token statistics play only a minor role. These findings support a modular view of multimodal AI in which learned or engineered front-ends extract structured visual languages that general-purpose LLMs can then reason over. Extending this approach beyond synthetic settings and developing robust mechanisms for acquiring such representations from pixels remain key directions for future work.

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